ORIGINAL RESEARCH/SCHOLARSHIP | Article in Press

Digitalisation, Use of Scientific Publications, and Research Output: Cross-Country Evidence

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Received: 27 September 2024 / Accepted: 17 September 2026
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Abstract
Access to scientific publications is essential for scientific progress, yet access remains uneven across countries due to differences in institutional subscriptions and digital infrastructure. This paper examines how digitalisation is associated with the use of accessible scientific publications, as proxied by Sci-Hub downloads, and scientific productivity across countries. Using Sci-Hub download data, we capture cross-country patterns in the use of accessible scientific knowledge. The results suggest that higher levels of digitalisation are associated with greater use of accessible scientific publications and higher levels of scientific output. Furthermore, the association between individual-level digitalisation and publication use is stronger in low- and lower-middle-income countries than in upper-middle- and high-income countries, whereas the corresponding income-group difference for business-level digitalisation is not statistically significant. These findings highlight the potential role of digitalisation in shaping global patterns of access to scientific knowledge and scientific production.

Keywords Digitalisation · Use of scientific publications · Scientific output · Inequality in science

Introduction

Access to scientific publications is important for knowledge creation and scientific productivity (Nascimento & Pólvora, 2018; Mueller-Langer et al., 2020; Hu et al., 2022). Open access has expanded the availability of scholarly knowledge, while alternative channels have also emerged beyond traditional subscription barriers (Greshake, 2017; Zhang et al., 2021). Against this background, this study examines how digitalisation is associated with the use of scientific publications, as proxied by Sci-Hub downloads, and with national scientific output. It also examines whether the association between digitalisation and publication use differs across countries at different income levels.

Institutional mandates and platform-based services provide several channels for accessing scientific publications. Examples include public access policies such as the National Institutes of Health Public Access Policy, repositories such as PubMed Central and arXiv, the Directory of Open Access Journals, open access journals such as PLOS, academic networking platforms such as ResearchGate, and discovery tools such as Google Scholar.

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https://doi.org/10.1007/s11948-026-00633-9

Published online: 12 October 2026

However, the availability of formal access channels varies across countries because of differences in funding support and digital infrastructure. Researchers in many regions may therefore continue to face barriers to accessing scientific publications. In this context, Sci-Hub has emerged as an alternative channel for accessing scholarly literature. It operates as a shadow library that provides access to paywalled articles outside conventional subscription channels.

Sci-Hub remains controversial because its provision of paywalled articles raises concerns about copyright and intellectual property protection (Greco, 2023). Nevertheless, studies document its widespread use across different regions, including at well-resourced universities in the United States and Europe (Bohannon, 2016; Geng et al., 2022; Pastor-Ramon et al., 2023). Recent web-traffic evidence also suggests substantial global engagement with Sci-Hub (Arshad & Maria, 2025).

Although Sci-Hub is not part of the formal open access infrastructure, its large-scale download data provide useful insights into cross-country patterns of article retrieval. Previous studies have used these data to examine unmet demand for paywalled publications and cross-country differences in access to scientific literature (Bohannon, 2016; Himmelstein et al., 2018). Compared with repositories and platforms such as PubMed Central, arXiv, ResearchGate, and Google Scholar, the available Sci-Hub logs provide a comparatively unified source of observed download behaviour across countries and a broad corpus of publications. Accordingly, this study treats Sci-Hub downloads as a proxy for one observable dimension of cross-national publication use, rather than as a measure of formal open access. Download records capture article retrieval but do not show whether the downloaded publications were fully read or applied. The study does not evaluate the legality of Sci-Hub.

Literature Review

Digitalisation and the Use of Scientific Publications

Digitalisation refers to the development and adoption of information and communication technologies (ICT) that enable the creation, access, and dissemination of digital information. It can facilitate access to and use of scientific publications by improving digital connectivity and information flows. Existing studies associate digitalisation with knowledge spillovers, knowledge diffusion, and improved access to knowledge resources (Batabyal & Nijkamp, 2016; Semenikhina et al., 2020; Deng et al., 2023).

International and national initiatives illustrate efforts to broaden access to scientific publications through digital infrastructure and coordinated policies. The Health InterNetwork Access to Research Initiative (HINARI), established by the World Health Organization together with major publishers, is one example (Aronson & Long, 2003). Other examples include the Network of Collaboration between Europe and Latin-American and the Caribbean Countries (NECOBELAC) and National Research and Education Networks (NRENs) in Mozambique (De Castro et al., 2012; Mulhanga et al., 2016).

Scientists working at the research frontier benefit from accessing and exchanging information, sharing ideas, and receiving feedback from their peers. The academic community has therefore emphasized the importance of channels and platforms that support scientific communication. Initiatives that reduce the cost of accessing knowledge, including open access, can broaden access to scientific publications (Eysenbach, 2006; Zhang et al., 2021). However, barriers remain, and limited digital infrastructure may restrict the accessibility and diffusion of scientific knowledge (Evans & Reimer, 2009; Biljon et al., 2017; Zhang et al., 2021). Improvements in digital infrastructure and communication systems may therefore broaden access to scientific publications, strengthen connectivity among researchers, and support knowledge exchange.

Digitalisation has been widely discussed as an important element in the development of modern science. Its role in scientific activity can be understood from several perspectives. First, digital technologies may improve research workflows and increase the efficiency of scientific processes (Hilbert, 2014; Yu et al., 2021). As a result, the development of digital technologies has been associated with changes in how scientific research is conducted.

Several studies examining digitalisation in science focus on its association with scientists' research output at the individual level (Yu et al., 2021). For example, using survey data from 338 oceanographic scientists, Hesse et al. (1993) found that internet use was positively associated with scientific outcomes. Similarly, using data from the Survey of Doctorate Recipients, Winkler et al. (2010) reported a positive association between information technology use and scientists' research output.

Digitalisation has also created additional opportunities for information sharing and knowledge use (Newman et al., 2012; Friesike et al., 2015; Fernandez-Patron & Hardy, 2018). It is widely recognized that knowledge is often protected by copyright law. Modern copyright systems aim to encourage the creation of new works by strengthening copyright protection. However, some studies argue that the expansion of copyright protection may also create barriers to knowledge access and reuse. Excessive copyright protection has been discussed as a potential constraint on innovation and knowledge diffusion (Ray et al., 2009; Heald, 2014).

At the same time, digital technologies have been associated with changes in traditional copyright-protected markets. Previous studies suggest that digitalisation may reduce the cost of distributing knowledge and creative content, thereby enabling new forms of information sharing and cultural production (Waldfogel, 2017; Wittel, 2013; Peukert, 2019). Some legal scholars also suggest that limited forms of digital piracy may be viewed more positively when they are used for non-commercial purposes or when they expand access to knowledge (Ciesielska & Jemielniak, 2022; Kleio & Alejo José, 2016). In the context of scientific communication, certain digital platforms that provide access to scientific literature, such as Sci-Hub, remain controversial in terms of legality. However, several studies suggest that these platforms have been widely used by researchers and may expand access to scientific literature (Correa et al., 2022; Buehling et al., 2022).

Previous studies also suggest that digitalisation is associated with the growing use of freely accessible knowledge (Ramachandran et al., 2021) by enabling new forms of resource sharing, collaboration, and outreach (Toelch & Ostwald, 2018). Increasing levels of digitalisation may also expand access to scientific knowledge for researchers in many regions of the world, particularly in developing countries (Arunachalam, 2003; Nagaraj et al., 2020). Given the potential role of digitalisation in scientific communication, Biljon et al. (2017) highlighted the importance of digital platforms for research collaboration and discussed ways to improve digital infrastructure to support knowledge sharing.

Digitalisation and Scientific Output Through the Use of Scientific Publications

Studies of freely accessible scientific knowledge have largely focused on knowledge dissemination and the citation advantage associated with open access publications (Eysenbach, 2006; Zhang et al., 2021). Less attention has been paid to whether the use of scientific publications is associated with the production of new knowledge, as existing research has mainly examined access to and dissemination of scientific information (Davis & Walters, 2011). This gap highlights the need to examine the relationship between publication use and national scientific output at the country level.

Access to scientific publications expands the knowledge inputs available to researchers and may support the production of new knowledge. Previous studies associate the acquisition and use of external knowledge with research and innovation outcomes (Segarra-Ciprés et al., 2014; Limaj & Bernroider, 2019; Ngoc Thang & Anh Tuan, 2020; Xie et al., 2020).

Although many researchers argue that extensive knowledge sharing is an essential component of scientific progress (McKiernan et al., 2016; Sengupta & Shanahan, 2017), difficulties in accessing scientific literature have historically limited the development of science (Correa et al., 2022; Tennant et al., 2016). Other studies highlight the communication advantages associated with reducing the cost of knowledge sharing and report benefits in terms of citations, social media visibility, and usage (Evans & Reimer, 2009; Correa et al., 2022; Zhang et al., 2021). Björk (2017) further suggested that access to freely available scientific papers may be positively associated with the production of new knowledge.

Greater openness in access to scientific knowledge, including access through alternative channels, may improve the efficiency of scientific research by facilitating resource sharing and supporting more sustainable research activities (Grahe et al., 2020; Nagaraj et al., 2020). Researchers who gain access to scientific resources that were previously unavailable to them may have greater opportunities to produce new knowledge. Using data from 36,202 journal articles published by authors affiliated with 2,490 research institutions, Mueller-Langer et al. (2020) showed that improved online access to research materials is associated with increased research activity in developing countries. Using a Sci-Hub access dataset, Buehling et al. (2022) found that free access to academic knowledge is associated with greater representation of authors from developing countries. Other empirical studies also suggest that reducing the cost of accessing knowledge or removing barriers to knowledge access, such as licensing patents (Watzinger et al., 2020), weakening copyright restrictions (Biasi & Moser, 2021), and reducing the cost of accessing scientific data (Nagaraj et al., 2020), may create a more supportive environment for scientific communication and knowledge creation.

Digitalisation and Inequality in Science

Digitalisation has become an important component of contemporary scientific communication. Previous studies associate digitalisation with improvements in the scientific environment and researchers' productivity (Ribeiro et al., 2023; Zhang et al., 2025). Other studies have examined its relationship with national economic performance (Niebel, 2018). However, relatively few studies have examined the country-level relationships among digitalisation, the use of scientific publications, and national scientific output.

Facilitating equitable access to scientific knowledge and addressing inequality in science are important global concerns (Timmermann, 2014; Mormina, 2019). Open science initiatives may help reduce inequality by broadening access to freely available scientific resources, particularly for researchers in developing countries (Evans & Reimer, 2009; Nagaraj et al., 2020). However, expanding open science requires sustained and coordinated investment in digital infrastructure (UNESCO, 2021).

Digitalisation varies substantially across countries. Differences in digital infrastructure, often described as the digital divide, may limit scientific communication and the diffusion of scientific knowledge (Bezuidenhout et al., 2017; Forero-Pineda & Jaramillo-Salazar, 2002; Heeks, 2022). Researchers in developing countries have historically had more limited access to scientific information than those in developed countries (Molla, 2000; Arunachalam, 2003; Canessa et al., 2009). Consequently, making scientific resources openly available does not necessarily ensure equal access across countries.

Empirical findings on the relationship between the use of freely available scientific publications and global scientific participation remain mixed. Evans and Reimer (2009) suggest that open access is associated with broader global participation in science. In contrast, McCabe and Snyder (2011) found no significant regional differences in citation rates and argued that the benefits of online access may be smaller for scholars in developing countries than for those in Western countries. Other studies indicate that these benefits depend partly on technological infrastructure and that low-income countries may continue to show relatively low levels of article use despite policies that expand access (Nwagwu, 2016; Zhang et al., 2021).

Digital technologies may help narrow information gaps between developed and developing countries, but their effects vary across institutional and regional contexts (Arunachalam, 2003; Bezuidenhout et al., 2017). Because lower-income countries often have more limited digital and research infrastructure, improvements in digitalisation may be particularly important for expanding their access to scientific publications (Heeks, 2022).

Research Questions

Based on the preceding literature, this study addresses the following research questions:

RQ1: Is digitalisation associated with greater use of scientific publications, as proxied by Sci-Hub downloads?

RQ2: Is greater use of scientific publications associated with higher national scientific output?

RQ3: Is the association between digitalisation and the use of scientific publications, as proxied by Sci-Hub downloads, stronger in lower-income countries than in higher-income countries?

Figure 1 summarizes the relationships examined in the empirical analysis. Sci-Hub downloads are used as a proxy for publication use, while national scientific output is measured using publications indexed in the Web of Science. Download records provide an observable indicator of article retrieval but do not show whether the downloaded publications were fully read or applied. The figure also indicates that the association between digitalisation and publication use may differ across country income groups.

Methods and Data

Measurement

This study combines four datasets. The first consists of Sci-Hub download logs for 2017 and is used to measure country-level use of scientific publications. The second provides the 2016 Digital Adoption Index for businesses (BDAI) and the Digital Adoption Index for people (PDAI) and is used to measure digitalisation. The third contains country-level economic, educational, and demographic characteristics used as control variables. The fourth consists of Web of Science publication counts for 2018 and is used to measure national scientific output. The following subsections describe the measurement and sources of these variables.

Dependent Variable

The first dataset consists of Sci-Hub download logs for 2017. It represents one of the most comprehensive publicly available datasets providing large-scale information on cross-national downloads of scientific publications. The dataset was obtained from the Sci-Hub server logs shared through the @Sci-Hub Twitter account (Geng et al., 2022). It contains approximately 140 million download records between January 1, 2017 and December 31, 2017. Following previous studies, we use the number of Sci-Hub downloads at the country level as a proxy for the use of scientific publications.

Sci-Hub was founded in 2011 by the Kazakh neuroscientist Alexandra Elbakyan and operates as a platform that provides access to a large number of scholarly publications (James, 2020; Himmelstein et al., 2018; Bohannon, 2016; Geng et al., 2022). Although the dataset is limited to 2017, it remains useful for this study because it provides a large-scale and comparable country-level record of observed article-download behavior across a broad cross-section of countries and therefore offers a useful basis for examining cross-country differences in the use of scientific publications (Gao et al., 2024; Kjellström, 2025; Osman & Ahmed, 2024).

We cleaned the dataset by removing records associated with automated downloading behavior following the procedure described in Geng et al. (2022). Automated downloading activity was identified based on the temporal patterns of downloads associated with user IDs and IP addresses. Three criteria were applied. First, user IDs were flagged as potential automated downloaders if more than 50% of the time intervals between consecutive downloads were shorter than 10 s. Second, if a user ID downloaded more than 60 articles within one hour, the activity was considered likely to be generated by automated scripts or crawlers. Third, IP addresses that showed continuous downloading activity for more than 18 h without a break of at least 6 h were treated as potential automated or proxy downloads. User IDs and IP addresses that met these criteria were removed in order to retain download

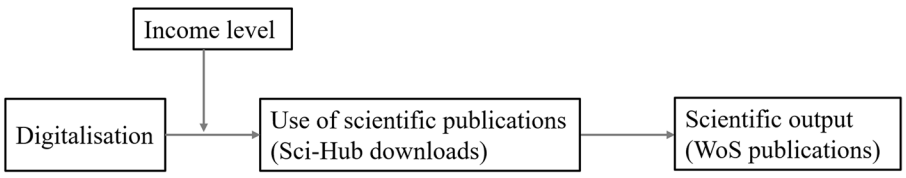


Fig. 1 Conceptual framework of the study

patterns that more closely resemble human behavior. After removing these records and excluding observations with incomplete fields, we standardized the names of countries and regions. The final dataset contains 74,742,113 valid download observations.

The empirical design follows a temporal sequence in which digitalisation is measured before Sci-Hub download activity, and Sci-Hub download activity is measured before scientific output. This structure is intended to align the timing of the variables with the hypothesized relationships examined in the study.

Independent Variable

The second dataset contains measures of digitalisation. The sample is determined by data availability. The key variables of interest are the Digital Adoption Index for businesses (BDAI) and the Digital Adoption Index for people (PDAI). BDAI is constructed as the average of four indicators: the percentage of businesses with websites, the number of secure servers, download speed, and 3G coverage. PDAI is calculated as the average of two indicators: mobile phone access at home and internet access at home. These indicators capture digital adoption across two dimensions of the economy, namely businesses and individuals. The data are obtained from the World Bank and the Gallup World Poll. Although measures are available for 2014 and 2016, this study uses the 2016 values (Güvercin, 2022). These indicators precede the Sci-Hub download data used in this study and therefore reflect the level of digitalisation before the observed use of scientific publications.

Control Variables

The third dataset describes country-level characteristics. We include several control variables that may be associated with Sci-Hub downloads. These variables include GDP per capita growth, government expenditure on education, and mean years of schooling. Following previous studies (Gao et al., 2017; Whetsell et al., 2021; Brzezinski, 2022; Güvercin, 2022), these variables are used to account for differences in economic development, education systems, and human capital across countries.

We also include country size variables, namely land area and population, in Eq. (1) following Güvercin (2022). These variables are included to account for differences in the potential scale of users across countries. The definitions and sources of all variables are reported in Table 1. Summary statistics and a pairwise correlation matrix for the variables used in Eq. (1) are presented in Table 2.

Table 1 Variable definitions

Variables	Variable name	Description	Sources	Time
Use of scientific publications	*Sci-Hub*	Number of Sci-Hub downloads per country	Sci-Hub download logs	2017
Digital Adoption Index (Business)	*BDAI*	Average index of: percentage of businesses with websites, number of secure servers, download speed, and 3G coverage	World Bank & Gallup World Poll	2016
Digital Adoption Index (People)	*PDAI*	Average index of: mobile access at home and internet access at home	World Bank & Gallup World Poll	2016
GDP per capita growth	*Gdpg*	Annual percentage growth rate of GDP per capita based on constant local currency	World Bank	2017
Government expenditure on education	*Edu*	Percentage of total government expenditure allocated to education	World Bank	2017
Mean years of schooling	*Schooling*	Average number of years of education received by people aged 25 and above	UNDP	2017
Population	*Pop*	Total population	World Bank	2017
Area	*Area*	Total land area of the country	World Bank	2017
Scientific output	*Output*	Number of publications (articles and reviews) per country	Web of Science	2018

Table 2 The correlation matrix and descriptive statistics of applied variables (N=139)

	Variable	1	2	3	4	5	6	7	8
1	Sci-Hub	1.00							
2	*PDAI*	0.17	1.00						
3	*BDAI*	0.14	0.90	1.00					
4	*Gdpg*	0.12	0.07	0.10	1.00				
5	*Edu*	-0.06	-0.29	-0.30	0.14	1.00			
6	*Schooling*	0.10	0.82	0.84	0.14	-0.27	1.00		
7	*Pop*	0.86	-0.03	-0.07	0.14	-0.03	-0.07	1.00	
8	*Area*	0.61	0.05	0.01	-0.01	0.00	0.04	0.52	1.00
	Mean	418785.57	0.47	0.59	2.01	14.59	8.84	34236031.92	640429.45
	S.D.	1204570.15	0.24	0.20	2.83	4.70	3.09	122858567.55	1873995.30
	Min	12.00	0.05	0.13	-7.04	5.12	1.50	71460.00	300.00
	Max	11316634.00	0.91	0.97	7.94	30.15	14.10	1396215000.00	16376870.00

Note: N=139 countries. Sci-Hub denotes the number of Sci-Hub downloads per country. PDAI refers to the Digital Adoption Index (People), and BDAI refers to the Digital Adoption Index (Business). Gdpg denotes GDP per capita growth. Edu refers to government expenditure on education. Schooling denotes mean years of schooling. Pop represents total population, and Area denotes the total land area of a country

Scientific Output

The fourth dataset measures countries' scientific output. We use the Web of Science database to collect publication data for the document types "Article" and "Review" in journals indexed in SCI Expanded and SSCI for the year 2018.

Scientific output is measured using publication counts for 2018. This timing allows us to examine the association between the use of scientific publications in 2017 and subsequent research production. Accessing and reading scientific literature typically precedes the research process, including idea development, experimentation, manuscript preparation, and publication. Therefore, download activity in 2017 may be associated with research outputs produced in subsequent periods, including 2018. This temporal ordering helps reduce concerns about simultaneity, although it does not eliminate the possibility that pre-existing scientific activity or omitted factors are associated with both download activity and subsequent scientific output.

Empirical Strategy

Sci-Hub download counts are non-negative integers. To reduce the influence of extreme values and distributional skewness, we use the logarithm of Sci-Hub downloads, denoted as lnSci-Hub, as the dependent variable. Ordinary least squares regression is employed to examine the association between digitalisation and publication use. The following models are estimated:

$$lnSci-Hubi = \beta_0 + \beta_1 PDAIi + \beta_2 Gdpgi + \beta_3 Edui + \beta_4 Schoolingi + \beta_5 Popi + \beta_6 Areai + ui \quad (1)$$

$$lnSci-Hubi = \beta_0 + \beta_1 BDAIi + \beta_2 Gdpgi + \beta_3 Edui + \beta_4 Schoolingi + \beta_5 Popi + \beta_6 Areai + ui \quad (2)$$

where ui denotes the error term. The coefficient β_1 captures the association between digitalisation and publication use, as proxied by Sci-Hub downloads.

To further examine whether publication use represents one pathway through which digitalisation is associated with scientific output, we conduct a mediation analysis using a bootstrap procedure with 500 replications. The mediation framework estimates the indirect association between digitalisation and scientific output through publication use.

In the two mediation models, PDAI and BDAI are used separately as independent variables. The logarithm of Sci-Hub downloads (lnSci-Hub), which represents the use of scientific publications, is used as the mediating variable, and the logarithm of scientific output (lnOutput) is used as the dependent variable. GDP per capita growth, population, and area are included as control variables. The sample size used in the mediation analysis differs from that used in the baseline regressions. Some control variables in the baseline models contain missing observations for several countries, which limits the baseline analysis to 139 countries. In the mediation models, fewer control variables are included. This allows additional observations to be retained, resulting in a larger sample of 164 countries. In addition, the more parsimonious specification helps mitigate potential multicollinearity among explanatory variables. Two mediation models are estimated. The first model uses the Digital Adoption Index (People) (PDAI) as the independent variable, while the second model uses the Digital Adoption Index (Business) (BDAI). In each case, the first equation estimates the total association between digitalisation and scientific output, the second equation estimates the association between digitalisation and lnSci-Hub, and the third equation estimates the direct association between digitalisation and scientific output while including lnSci-Hub as the mediating variable.

Mediation Model Using PDAI

First, we estimate the overall association between digitalisation and scientific output:

$$lnOutputi = \beta_0 + \beta_1 PDAIi + \beta_2 Gdpgi + \beta_3 Popi + \beta_4 Areai + ui \quad (3)$$

Second, we estimate the association between digitalisation and lnSci-Hub, the mediating variable:

$$lnSci-Hubi = \alpha + \beta_1 PDAIi + \beta_2 Gdpgi + \beta_3 Popi + \beta_4 Areai + ui \quad (4)$$

Third, we estimate the association between digitalisation and scientific output while including lnSci-Hub as the mediating variable:

$$lnOutputi = \beta_0{}' + \beta_1 PDAIi + \beta_2 lnSci-Hubi + \beta_3 Gdpgi + \beta_4 Popi + \beta_5 Areai + ui \quad (5)$$

Mediation Model Using BDAI

We repeat the same procedure using BDAI as the independent variable and lnSci-Hub as the mediating variable:

$$lnOutputi = \beta_0 + \beta_1 BDAIi + \beta_2 Gdpgi + \beta_3 Popi + \beta_4 Areai + ui \quad (6)$$

$$lnSci-Hubi = \alpha + \beta_1 BDAIi + \beta_2 Gdpgi + \beta_3 Popi + \beta_4 Areai + ui \quad (7)$$

$$lnOutputi = \beta_0{}' + \beta_1 BDAIi + \beta_2 lnSci-Hubi + \beta_3 Gdpgi + \beta_4 Popi + \beta_5 Areai + ui \quad (8)$$

Results

In this section, we present the empirical results. First, we provide an overview of the main variables used in the analysis. After removing countries with missing data, the dataset includes 139 observations, where each observation represents a country with several characteristics, including PDAI, BDAI, Gdpg, Edu, Schooling, Pop, and

Area. Following the World Bank income classification, these countries are grouped into two categories: Low and Lower Middle-Income Countries (LMIC), which include low income and lower middle-income countries, and Upper Middle- and High-Income Countries (MHIC), which include upper middle income and high-income countries.

The World Bank (2018) reports that the Digital Adoption Index was highly correlated with per capita income in both 2014 and 2016 and that all ten countries with the highest DAI scores in 2016 were high-income countries. National scientific activity is also associated with economic scale, as Rodríguez-Navarro and Brito (2022) find a strong relationship between national GDP and the number of scientific publications. These findings provide context for the relationships among digital adoption, Sci-Hub downloads, and scientific output examined in Fig. 2.

Next, we examine the relationships between digitalisation, Sci-Hub downloads, and scientific output. The scatterplots in Fig. 2 provide an initial visual illustration of the relationships between these variables. The left panel shows the relationship between lnSci-Hub and the Digital Adoption Index (People), while the right panel shows the relationship between lnSci-Hub and the Digital Adoption Index (Business). In the figure, circles represent Low and Lower Middle-Income Countries, while squares represent Upper Middle- and High-Income Countries. The size of each marker reflects the logarithm of scientific output (lnOutput) for each country. Larger markers therefore indicate higher levels of scientific output. Fitted lines are also presented for the two groups of countries.

The scatterplots suggest a positive association between the digital adoption indices and ln(Sci-Hub downloads). The slope of the fitted line appears steeper for Low and Lower Middle-Income Countries than for Upper Middle- and High-Income Countries. This pattern is particularly visible in the left panel. In addition, observations with higher levels of digital adoption and higher ln(Sci-Hub downloads) tend to have larger marker sizes, which correspond to higher levels of scientific output. These patterns provide preliminary descriptive evidence that digitalisation may be associated with greater use of scientific publications and higher levels of scientific output.

Baseline Results

To examine the robustness of the baseline results, we estimate several alternative model specifications. The main analysis uses OLS regression with the logarithm of Sci-Hub downloads as the dependent variable. We use this as the baseline model because the logarithmic transformation reduces the influence of extreme skewness in the distribution and allows a straightforward interpretation of the estimated association.

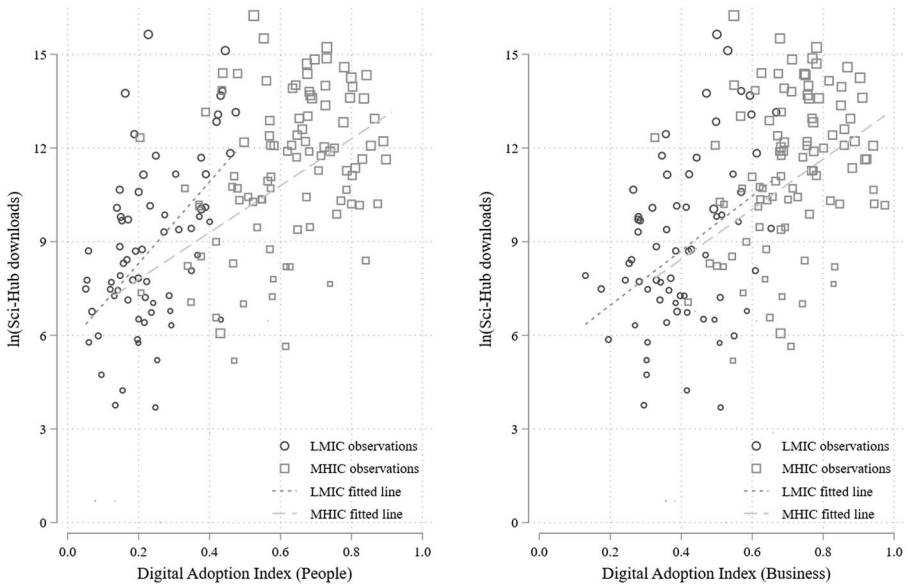


Fig. 2 Scatterplots of ln(Sci-Hub downloads) and the Digital Adoption Index for People (left panel) and Business (right panel)

However, the original Sci-Hub download variable is a non-negative count variable. Therefore, we additionally estimate Poisson and Negative Binomial models, which are commonly used for count data, as alternative specifications. In addition, we estimate a specification in which Sci-Hub downloads are measured in millions rather than transformed logarithmically. Comparing these different models allows us to examine whether the estimated association between digitalisation and Sci-Hub downloads remains consistent across different estimation approaches.

Table 3 reports the results of the regression models presented in Eqs. (1) and (2). Columns (1) to (5) in Panel A present the results using PDAI, while columns (6) to (10) in Panel B present the results using BDAI. The results in columns (1) and (6) show that both the Digital Adoption Index (People) and the Digital Adoption Index

Table 3 Regression results for digitalisation and Sci-Hub downloads

Panel A: Digitalisation (People) and Sci-Hub downloads					
	OLS	**Nbreg**	**Poisson**	**Sci-Hub downloads (in millions)**	**No Schooling**
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
PDAI	5.29***	7.35***	2.51**	0.94***	5.70***
	(1.66)	(1.33)	(1.21)	(0.30)	(0.99)
Gdpg	0.24***	0.12	0.01	-0.00	0.24***
	(0.05)	(0.07)	(0.05)	(0.01)	(0.05)
Edu	0.01	0.04	0.02	0.00	0.01
	(0.04)	(0.04)	(0.04)	(0.01)	(0.04)
Schooling	0.04	0.02	-0.00	0.00	
	(0.10)	(0.09)	(0.12)	(0.02)	
Pop	0.00	0.00*	0.00***	0.00***	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Area	0.00***	0.00	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Constant	6.41***	6.30***	10.86***	-0.43**	6.58***
	(1.01)	(0.92)	(0.90)	(0.21)	(0.94)
Obs	139	139	139	139	139
ll	-	-1719.38	-44364834.74	-	-
aic	-	3454.76	88729681.48	-	-
bic	-	3478.24	88729699.09	-	-
R^2	0.39	-	-	0.79	0.40
Panel B: Digitalisation (Business) and Sci-Hub downloads					
	OLS	**Nbreg**	**Poisson**	**Sci-Hub downloads (in millions)**	**No Schooling**
	(6)	**(7)**	**(8)**	**(9)**	**(10)**
BDAI	5.47***	6.46***	4.29**	1.32***	6.45***
	(1.80)	(1.37)	(2.18)	(0.46)	(0.99)
Gdpg	0.23***	0.16**	0.01	-0.01	0.23***
	(0.05)	(0.06)	(0.04)	(0.01)	(0.05)
Edu	0.01	0.02	0.02	0.01	0.01
	(0.04)	(0.03)	(0.04)	(0.01)	(0.04)
Schooling	0.08	0.12	-0.07	-0.01	
	(0.11)	(0.08)	(0.15)	(0.03)	
Pop	0.00	0.00	0.00***	0.00***	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Area	0.00***	0.00	0.00***	0.00**	0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Constant	5.33***	5.29***	10.00***	-0.68***	5.45***
	(1.06)	(0.83)	(0.71)	(0.21)	(1.04)
Obs	139	139	139	139	139
ll	-	-1722.52	-42735901.90	-	-
aic	-	3461.05	85471815.80	-	-
bic	-	3484.52	85471833.41	-	-
R^2	0.37	-	-	0.79	0.38

Note: * $p<0.1$, ** $p<0.05$, *** $p<0.01$. OLS refers to ordinary least squares regression using lnSci-Hub as the dependent variable. Nbreg refers to the Negative Binomial regression model, and Poisson refers to the Poisson regression model estimated using the original count of Sci-Hub downloads. "Sci-Hub downloads (in millions)" uses Sci-Hub downloads measured in millions as the dependent variable. Standard errors are reported in parentheses

(Business) are positively and significantly associated with lnSci-Hub, after controlling for the country characteristics included in the model. The results reported in columns (2) to (4) and (7) to (9), which correspond to the Poisson model, the Negative Binomial model, and the specification using Sci-Hub downloads measured in millions, are consistent with the OLS results shown in columns (1) and (6).

Because Schooling is highly correlated with PDAI and BDAI, we also estimate models that exclude Schooling. The corresponding results are reported in columns (5) and (10). The estimated coefficients for PDAI and BDAI remain positive and statistically significant in these specifications. Overall, the consistency of the results across different model specifications suggests that the association between digitalisation and Sci-Hub downloads is robust.

Heterogeneous Effects Among Countries at Different Income Levels

Before estimating the relationship between digitalisation and Sci-Hub downloads across countries with different income levels, we classify the sample into two subgroups based on income level: Low and Lower Middle-Income Countries (LMIC) and Upper Middle- and High-Income Countries (MHIC). Table 4 reports the regression results for the two subgroups. The coefficients of PDAI and BDAI for LMIC (11.65 and 7.43) are positive and statistically significant at the 5% level. These coefficients are larger than those for MHIC (3.73 and 4.11), which are not statistically significant. This pattern suggests that the association between digitalisation and lnSci-Hub may be stronger in LMIC than in MHIC.

To examine whether the coefficients differ significantly between the two groups (LMIC and MHIC countries), we conduct statistical tests based on the seemingly unrelated regression (SUR) framework. SUR is an econometric method that allows multiple regression equations to be estimated jointly while accounting for potential correlations among the error terms across equations (Zellner, 1962). By estimating the equations jointly, SUR enables formal cross-equation tests of whether corresponding coefficients differ significantly between groups. This approach allows formal tests of whether estimated coefficients differ between groups.

The p-values are obtained under the null hypothesis that the coefficients are equal across the two groups. The alternative hypothesis assumes that the coefficient for LMIC is larger than that for MHIC. For example, the p-value of 0.0312 for PDAI indicates that the coefficient for LMIC is larger than that for MHIC at the 5% significance level. These results suggest that the association between PDAI and Sci-Hub downloads is stronger in LMIC than in MHIC. In contrast, the difference between LMIC and MHIC for BDAI is not statistically significant. Overall, these findings suggest that digitalisation at the individual level is more strongly associated with the use of scientific publications in LMIC than in MHIC.

Mediation Effect

Table 5 presents the results of the mediation analysis based on Eq. (3) to (8). The table reports the estimated direct and indirect associations between digitalisation and scientific output through the use of scientific publications.

Table 4 OLS regression results by income-level subgroups (PDAI and BDAI models)

	PDAI		BDAI	
	LMIC	MHIC	LMIC	MHIC
Digitalisation	11.65***	3.73	7.43***	4.11
	(2.20)	(2.33)	(2.42)	(2.94)
Observations	55	84	55	84
R^2	0.69	0.22	0.59	0.21
Controls	Yes	Yes	Yes	Yes
p-values	0.0312**		0.3360	

Note: * $p<0.1$, ** $p<0.05$, *** $p<0.01$

Table 5 Mediation analysis results

	Total	LMIC	MHIC
Indirect effect (PDAI)	3.903***	6.468***	3.411***
	(0.512)	(1.383)	(0.918)
Direct effect (PDAI)	1.634***	-2.329*	3.929***
	(0.629)	(1.406)	(0.778)
Indirect effect (BDAI)	4.458***	3.806***	3.822***
	(0.637)	(1.243)	(1.196)
Direct effect (BDAI)	1.776**	-1.115	4.750***
	(0.690)	(0.931)	(1.136)
Observations	164	68	96
Controls	Yes	Yes	Yes

Note: * $p<0.1$, ** $p<0.05$, *** $p<0.01$. Control variables include GDP per capita growth, population, and area

Results are reported for the full sample of countries as well as for two subsamples: Low and Lower Middle-Income Countries (LMIC) and Upper Middle- and High-Income Countries (MHIC). For each group, we report the mediation results for both the Digital Adoption Index (People) (PDAI) and the Digital Adoption Index (Business) (BDAI).

The results show that the estimated indirect effects are positive and statistically significant across all models. This pattern suggests that higher levels of digitalisation are associated with greater use of scientific publications, which in turn is associated with higher levels of scientific output.

For LMIC countries, the indirect effects remain positive and statistically significant, while the direct effects are negative and mixed in terms of statistical significance. Specifically, the direct effect is negative and statistically significant for the PDAI model, but negative and not statistically significant for the BDAI model. This pattern suggests that the association between digitalisation and scientific output in LMIC may operate mainly through the use of scientific publications, as proxied by Sci-Hub downloads. The opposite signs of the direct and indirect effects indicate that the estimated effects move in different directions.

Discussion

This study examines the relationships among digitalisation, the use of scientific publications, and national scientific output. The results suggest that higher levels of digitalisation are associated with greater use of scientific publications, as proxied by Sci-Hub downloads. Sci-Hub downloads are also positively associated with national scientific output. The association between individual-level digitalisation and publication use is stronger in low- and lower-middle-income countries than in upper-middle- and high-income countries, whereas the corresponding income-group difference for business-level digitalisation is not statistically significant. These findings contribute to the literature by providing country-level evidence on the relationships among digitalisation, publication use, and scientific output.

These findings are consistent with evidence that digital technologies and online access shape scientific communication and everyday scientific work (Evans, 2008; Ribeiro et al., 2023). In addition, this study provides further evidence that may have several policy implications. International cooperation may play an important role in improving the digital capacity of low- and lower-middle-income countries. Efforts to expand digital infrastructure may help improve access to scientific publications. In addition to expanding access to scientific publications, it may also be important to improve access to digital tools such as personal computers and reliable internet connectivity.

The mediation analysis further suggests that digitalisation is associated with scientific output partly through the use of scientific publications. In low- and lower-middle-income countries, the indirect effects remain positive and statistically significant, whereas the direct effect is negative and statistically significant in the PDAI model but

negative and not statistically significant in the BDAI model. This pattern suggests that the association between digitalisation and scientific output in these countries may operate partly through the use of scientific publications, as proxied by Sci-Hub downloads. In contrast, in upper-middle- and high-income countries, both the direct and indirect effects remain positive. This result suggests that digitalisation may be associated with scientific output through publication use as well as through other possible channels. More broadly, these findings are consistent with the view that the effects of digitalisation vary across institutional and regional contexts (Tóth & Wang, 2026). Access to scientific knowledge resources may therefore be particularly relevant in low- and lower-middle-income countries.

The stronger association between PDAI and publication use in LMIC may reflect the importance of individual-level digital access in lower-income settings. This interpretation is consistent with evidence that mobile information and communication technologies can expand rapidly in emerging economies because of their relatively lower infrastructure requirements (Kim et al., 2021). Such forms of asset-light digitalisation may help broaden access to scientific knowledge resources, especially in developing countries.

In addition to investments in hardware and devices, improving the effectiveness of digitalisation may require cooperation between governments and the academic community. When developing digital strategies at national and regional levels, policymakers may consider expanding access to online scientific resources.

At the same time, digitalisation involves multiple processes, including classification, recognition, and computation, which require specific capabilities (Morzy, 2015; Zhang et al., 2025). Therefore, improving the use of scientific publications may also depend on the skills required to manage digital technologies and information systems. These capabilities are related to digital management and knowledge sharing practices. Relevant skills may therefore be incorporated into higher education curricula.

Limitations

We acknowledge that this study has several limitations. First, Sci-Hub is not a conventional open access repository. It is commonly described as a shadow library because it provides unauthorized access to paywalled scholarly publications, which raises intellectual property and governance concerns (Greco, 2023; Himmelstein et al., 2018). In addition, the Sci-Hub dataset records article retrieval through only one platform and does not indicate whether downloaded publications were fully read, understood, or applied. It also does not capture other access channels, including institutional subscriptions, formal open access repositories, and academic networking platforms. Accordingly, Sci-Hub downloads should be interpreted as one observable proxy for country-level publication use rather than a comprehensive measure of access to or use of scientific publications.

Second, potential endogeneity may exist in the relationships among digitalisation, publication use, and scientific output. For example, countries with higher levels of scientific activity may have greater demand for scholarly literature, which may be associated with higher download activity. To partly address concerns about temporal ordering, the mediation analysis uses scientific output measured after the Sci-Hub download data. However, this temporal ordering does not eliminate the possibility of reverse relationships or omitted-variable bias. Although instrumental variable approaches could potentially provide additional evidence, identifying valid cross-country instruments that influence digitalisation while remaining unrelated to publication use and scientific output is challenging given current data availability. Therefore, the findings should be interpreted primarily as evidence of associations rather than causal relationships. Future research may use alternative identification strategies or new datasets to further examine the relationships among digitalisation, publication use, and scientific output.

Conclusion

This study provides country-level evidence on the relationships among digitalisation, the use of scientific publications, and national scientific output. Both individual-level and business-level digitalisation are positively associated with Sci-Hub downloads, which are used as a proxy for publication use. Sci-Hub downloads are also positively associated with subsequent scientific output. The association between individual-level digitalisation and Sci-Hub downloads is stronger in low- and lower-middle-income countries than in upper-middle- and high-income countries, whereas the corresponding income-group difference for business-level digitalisation is not statistically significant. The mediation results further show positive and statistically significant indirect associations through Sci-Hub downloads.

Because the analysis uses observational cross-country data and Sci-Hub downloads capture only one channel of publication access, the findings should be interpreted as associations rather than causal effects. However, the results highlight the importance of strengthening digital infrastructure, digital skills, and lawful and sustainable access to scholarly literature. These measures may be particularly relevant for supporting more inclusive participation in scientific knowledge production in lower-income countries.

Acknowledgements The authors thank the associate editor Qin Zhu and two reviewers for their insightful comments and suggestions.

Declarations

Conflict of Interest We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work.

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References

Aronson, B., & Long, M. (2003). HINARI: Health InterNetwork Access to Research Initiative. *Serials*, *16*(1), 7–12. https://doi.org/10.1629/167

Arshad, A., & Maria, R. (2025). What do users prefer: Google Scholar or black open access? A comparative log analysis study. *Journal of Information Science*. https://doi.org/10.1177/01655515251330604

Arunachalam, S. (2003). Information for Research in Developing Countries—Information Technology, a Friend or Foe? *International Information & Library Review*, *35*(2–4), 133–147. https://doi.org/10.1080/10572317.2003.10762596

Batabyal, A. A., & Nijkamp, P. (2016). Digital technologies, knowledge spillovers, innovation policies, and economic growth in a creative region. *Economics of Innovation and New Technology*, *25*(5), 470–484. https://doi.org/10.1080/10438599.2015.1073485

Bezuidenhout, L. M., Leonelli, S., Kelly, A. H., & Rappert, B. (2017). Beyond the digital divide: Towards a situated approach to open data. *Science and Public Policy*, *44*(4), 464–475. https://doi.org/10.1093/scipol/scw036

Biasi, B., & Moser, P. (2021). Effects of Copyrights on Science: Evidence from the WWII Book Republication Program. *American Economic Journal: Microeconomics*, *13*(4), 218–260. https://doi.org/10.1257/mic.20190113

Björk, B. C. (2017). Open access to scientific articles: A review of benefits and challenges. *Internal and Emergency Medicine*, *12*(2), 247–253. https://doi.org/10.1007/s11739-017-1603-2

Bohannon, J. (2016). Who's downloading pirated papers? Everyone. *Science*, *352*(6285), 508–512. https://doi.org/10.1126/science.352.6285.508

Brzezinski, M. (2022). Does income redistribution impede innovation? *Research Policy*, *51*(10), 104603. https://doi.org/10.1016/j.respol.2022.104603
Buehling, K., Geissler, M., & Strecker, D. (2022). Free access to scientific literature and its influence on the publishing activity in developing countries: The effect of Sci-Hub in the field of mathematics. *Journal of the Association for Information Science and Technology*. https://doi.org/10.1002/asi.24636
Canessa, E., Zennaro, M., & Fonda, C. (2009). Supporting science in developing countries using open technologies. *European Journal of Physics*, *30*(3), 651–660. https://doi.org/10.1088/0143-0807/30/3/022
Ciesielska, M., & Jemielniak, D. (2022). Fairness in digital sharing legal professional attitudes toward digital piracy and digital commons. *Journal of the Association for Information Science and Technology*, *73*(7), 899–912. https://doi.org/10.1002/asi.24592
Correa, J. C., Laverde-Rojas, H., Tejada, J., & Marmolejo-Ramos, F. (2022). The Sci-Hub effect on papers' citations. *Scientometrics*, *127*(1), 99–126. https://doi.org/10.1007/s11192-020-03806-w
Davis, P. M., & Walters, W. H. (2011). The impact of free access to the scientific literature: A review of recent research. *Journal of the Medical Library Association: JMLA*, *99*(3), 208–217. https://doi.org/10.3163/1536-5050.99.3.008
De Castro, P., Marsili, D., Poltronieri, E., & Agudelo Calderon, C. (2012). Dissemination of public health information: Key tools utilised by the NECOBELAC network in Europe and Latin America. *Health Information and Libraries Journal*, *29*(2), 119–130. https://doi.org/10.1111/j.1471-1842.2012.00977.x
Deng, H., Duan, S. X., & Wibowo, S. (2023). Digital technology driven knowledge sharing for job performance. *Journal of Knowledge Management*, *27*(2), 404–425. https://doi.org/10.1108/JKM-08-2021-0637
Evans, J. A. (2008). Electronic Publication and the Narrowing of Science and Scholarship. *Science*, *321*(5887), 395–399. https://doi.org/10.1126/science.1150473
Evans, J. A., & Reimer, J. (2009). Open Access and Global Participation in Science. *Science*, *323*(5917), 1025–1025. https://doi.org/10.1126/science.1154562
Eysenbach, G. (2006). Citation Advantage of Open Access Articles. *PLOS Biology*, *4*(5), e157. https://doi.org/10.1371/journal.pbio.0040157
Fernandez-Patron, C., & Hardy, E. (2018). A New Science Publishing System for a Budding Science Publishing Crisis. *Science and Engineering Ethics*, *24*(2), 805–808. https://doi.org/10.1007/s11948-017-9902-6
Forero-Pineda, C., & Jaramillo-Salazar, H. (2002). The access of researchers from developing countries to international science and technology. *International Social Science Journal*, *54*(171), 129–140. https://doi.org/10.1111/1468-2451.00364
Friesike, S., Widenmayer, B., Gassmann, O., & Schildhauer, T. (2015). Opening science: Towards an agenda of open science in academia and industry. *The Journal of Technology Transfer*, *40*(4), 581–601. https://doi.org/10.1007/s10961-014-9375-6
Gao, Y., Zang, L., Roth, A., & Wang, P. (2017). Does democracy cause innovation? An empirical test of the popper hypothesis. *Research Policy*, *46*(7), 1272–1283. https://doi.org/10.1016/j.respol.2017.05.014
Gao, Z., Moskovkin, V. M., Gakhova, N. N., & Boichuk, I. V. (2024). Comparative analysis of downloads of articles by Chinese and Russian researchers using Sci-Hub. *Education and Information Technologies*. https://doi.org/10.1007/s10639-024-12507-3
Geng, Y., Cao, R., Han, X., Tian, W., Zhang, G., & Wang, X. (2022). Scientists are working overtime: When do scientists download scientific papers? *Scientometrics*, *127*(11), 6413–6429. https://doi.org/10.1007/s11192-022-04524-1
Grahe, J. E., Cuccolo, K., Leighton, D. C., & Cramblet Alvarez, L. D. (2020). Open Science Promotes Diverse, Just, and Sustainable Research and Educational Outcomes. *Psychology Learning & Teaching*, *19*(1), 5–20. https://doi.org/10.1177/1475725719869164
Greco, A. N. (2023). The Impact of Legal, Intellectual Property, and Copyright Infringement Issues: 2000–2022. In A. N. Greco (Ed.), *The Strategic Marketing of Science, Technology, and Medical Journals: A Business History of a Dynamic Marketplace, 2000–2020* (pp. 69–91). Springer International Publishing. https://doi.org/10.1007/978-3-031-31964-8_5
Greshake, B. (2017). Looking into Pandora's Box: The Content of Sci-Hub and its Usage (6:541). *F1000Research*. https://doi.org/10.12688/f1000research.11366.1
Güvercin, D. (2022). Digitalization and populism: Cross-country evidence. *Technology in Society*, *68*, 101802. https://doi.org/10.1016/j.techsoc.2021.101802
Heald, P. J. (2014). How Copyright Keeps Works Disappeared. *Journal of Empirical Legal Studies*, *11*(4), 829–866. https://doi.org/10.1111/jels.12057
Heeks, R. (2022). Digital inequality beyond the digital divide: Conceptualizing adverse digital incorporation in the global South. *Information Technology for Development*, *28*(4), 688–704. https://doi.org/10.1080/02681102.2022.2068492
Hesse, B. W., Sproull, L. S., Kiesler, S. B., & Walsh, J. P. (1993). Returns to science: Computer networks in oceanography. *Communications of the ACM*, *36*(8), 90–101. https://doi.org/10.1145/163381.163409
Hilbert, M. (2014). How much of the global information and communication explosion is driven by more, and how much by better technology? *Journal of the Association for Information Science and Technology*, *65*(4), 856–861. https://doi.org/10.1002/asi.23031

Himmelstein, D. S., Romero, A. R., Levernier, J. G., Munro, T. A., McLaughlin, S. R., Tzovaras, G., B., & Greene, C. S. (2018). Sci-Hub provides access to nearly all scholarly literature. *eLife*, *7*, e32822. https://doi.org/10.7554/eLife.32822
Hu, G., Ni, R., & Tang, L. (2022). Do international nonstop flights foster influential research? Evidence from Sino-US scientific collaboration. *Journal of Informetrics*, *16*(4), 101348. https://doi.org/10.1016/j.joi.2022.101348
James, J. E. (2020). Pirate open access as electronic civil disobedience: Is it ethical to breach the paywalls of monetized academic publishing? *Journal of the Association for Information Science and Technology*, *71*(12), 1500–1504. https://doi.org/10.1002/asi.24351
Kim, J., Park, J. C., & Komarek, T. (2021). The impact of Mobile ICT on national productivity in developed and developing countries. *Information & Management*, *58*(3), 103442. https://doi.org/10.1016/j.im.2021.103442
Kjellström, Z. (2025). Patterns of piracy: Sci-Hub and Sweden 2011–2018. *Internet Histories*, *9*(3), 252–273. https://doi.org/10.1080/24701475.2025.2482459
Kleio, A., & Alejo José, G., S. (Eds.). (2016). *The Challenges of Capitalism for Virtue Ethics and the Common Good: Interdisciplinary Perspectives*. Edward Elgar Publishing. https://www.elgaronline.com/display/edcoll/9781784717902/9781784717902.xml
Limaj, E., & Bernroider, E. W. N. (2019). The roles of absorptive capacity and cultural balance for exploratory and exploitative innovation in SMEs. *Journal of Business Research*, *94*, 137–153. https://doi.org/10.1016/j.jbusres.2017.10.052
McCabe, M. J., & Snyder, C. M. (2011). Did Online Access to Journals Change the Economics Literature? *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.1746243
McKiernan, E. C., Bourne, P. E., Brown, C. T., Buck, S., Kenall, A., Lin, J., McDougall, D., Nosek, B. A., Ram, K., Soderberg, C. K., Spies, J. R., Thaney, K., Updegrove, A., Woo, K. H., & Yarkoni, T. (2016). How open science helps researchers succeed. *eLife*, *5*, e16800. https://doi.org/10.7554/eLife.16800
Molla, A. (2000). Downloading or uploading? The information economy and Africa's current status. *Information Technology for Development*, *9*(3–4), 205–221. https://doi.org/10.1080/02681102.2000.9525333
Mormina, M. (2019). Science, Technology and Innovation as Social Goods for Development: Rethinking Research Capacity Building from Sen's Capabilities Approach. *Science and Engineering Ethics*, *25*(3), 671–692. https://doi.org/10.1007/s11948-018-0037-1
Morzy, M. (2015). ICT Services for open and citizen science. *World Wide Web-Internet and Web Information Systems*, *18*(4), 1147–1161. https://doi.org/10.1007/s11280-014-0303-3
Mueller-Langer, F., Scheufen, M., & Waelbroeck, P. (2020). Does online access promote research in developing countries? Empirical evidence from article-level data. *Research Policy*, *49*(2), 103886. https://doi.org/10.1016/j.respol.2019.103886
Mulhanga, M. M., Lima, S. R., & Massingue, V. (2016). An Evolutive Model for Open Science in Mozambique. *2016 15th Roedunet Conference - Networking in Education and Research*. http://www.webofscience.com/wos/woscc/full-record/WOS:000390713800012
Nagaraj, A., Shears, E., & de Vaan, M. (2020). Improving data access democratizes and diversifies science. *Proceedings of the National Academy of Sciences*, *117*(38), 23490–23498. https://doi.org/10.1073/pnas.2001682117
Nascimento, S., & Pólvora, A. (2018). Maker Cultures and the Prospects for Technological Action. *Science and Engineering Ethics*, *24*(3), 927–946. https://doi.org/10.1007/s11948-016-9796-8
Newman, G., Wiggins, A., Crall, A., Graham, E., Newman, S., & Crowston, K. (2012). The future of citizen science: Emerging technologies and shifting paradigms. *Frontiers in Ecology and the Environment*, *10*(6), 298–304. https://doi.org/10.1890/110294
Ngoc Thang, N., & Anh Tuan, P. (2020). Knowledge acquisition, knowledge management strategy and innovation: An empirical study of Vietnamese firms. *Cogent Business & Management*, *7*(1), 1786314. https://doi.org/10.1080/23311975.2020.1786314
Niebel, T. (2018). ICT and economic growth – Comparing developing, emerging and developed countries. *World Development*, *104*, 197–211. https://doi.org/10.1016/j.worlddev.2017.11.024
Nwagwu, W. E. (2016). Open Access in the Developing Regions: Situating the Altercations About Predatory Publishing. *Canadian Journal of Information and Library Science-Revue Canadienne Des Sciences De L Information Et De Bibliotheconomie*, *40*(1), 58–80.
Osman, I. R., & Ahmed, H. A. H. (2024). Bibliometric analysis of Sci-Hub downloads by Egyptian researchers. *Scientometrics*, *129*(4), 2371–2388. https://doi.org/10.1007/s11192-024-04951-2
Pastor-Ramon, E., Aguirre, O., García-Puente, M., Morán, M., J., & Herrera-Peco, I. (2023). Sci-Hub use among Spanish researchers: Enemy or a learning opportunity for libraries? *Journal of Information Science.* https://doi.org/10.1177/01655515221142432
Peukert, C. (2019). The next wave of digital technological change and the cultural industries. *Journal of Cultural Economics*, *43*(2), 189–210. https://doi.org/10.1007/s10824-018-9336-2
Ramachandran, R., Bugbee, K., & Murphy, K. (2021). From open data to open science. *Earth and Space Science*, *8*(5), Article e2020EA001562. https://doi.org/10.1029/2020EA001562
Ray, R. S., Sun, J., & Fan, Y. (2009). Does Copyright Law Promote Creativity—An Empirical Analysis of Copyright's Bounty. *Vanderbilt Law Review*, *62*(6), 1667–1746.

Ribeiro, B., Meckin, R., Balmer, A., & Shapira, P. (2023). The digitalisation paradox of everyday scientific labour: How mundane knowledge work is amplified and diversified in the biosciences. *Research Policy*, *52*(1), 104607. https://doi.org/10.1016/j.respol.2022.104607

Rodríguez-Navarro, A., & Brito, R. (2022). The link between countries' economic and scientific wealth has a complex dependence on technological activity and research policy. *Scientometrics*, *127*(5), 2871–2896. https://doi.org/10.1007/s11192-022-04313-w

Segarra-Ciprés, M., Roca-Puig, V., & Bou-Llusar, J. C. (2014). External knowledge acquisition and innovation output: An analysis of the moderating effect of internal knowledge transfer. *Knowledge Management Research & Practice*, *12*(2), 203–214. https://doi.org/10.1057/kmrp.2012.55

Semenikhina, O., Yurchenko, A. O., Sbruieva, A. A., Kuzminskyi, A., Kuchai, O., & Bida, O. A. (2020). The Open Digital Educational Resources in It-Technologies: Quantity Analysis. *Information Technologies and Learning Tools*, *75*(1), 331–348.

Sengupta, P., & Shanahan, M. C. (2017). Open Science, Public Engagement and the University (arXiv:1702.04855). *arXiv*. https://doi.org/10.48550/arXiv.1702.04855

Tennant, J. P., Waldner, F., Jacques, D. C., Masuzzo, P., Collister, L. B., & Hartgerink, C. H. J. (2016). The academic, economic and societal impacts of Open Access: An evidence-based review. *F1000Research, 5,* Article 632. https://doi.org/10.12688/f1000research.8460.3

Timmermann, C. (2014). Sharing in or Benefiting from Scientific Advancement? *Science and Engineering Ethics*, *20*(1), 111–133. https://doi.org/10.1007/s11948-013-9438-3

Toelch, U., & Ostwald, D. (2018). Digital open science-Teaching digital tools for reproducible and transparent research. *Plos Biology*, *16*(7), e2006022. https://doi.org/10.1371/journal.pbio.2006022

Tóth, Z. B., & Wang, L. (2026). Digitalization and gender equality in the European Union: A comparative analysis of Eastern and Western member states. *Technology in Society*, *86*, 103297. https://doi.org/10.1016/j.techsoc.2026.103297

UNESCO. (2021). *UNESCO Recommendation on Open Science (Programme and Meeting Document SC-PCB-SPP/2021/OS/UROS)*. UNESCO. https://unesdoc.unesco.org/ark:/48223/pf0000379949.locale=en

van Biljon, J., Marais, M., & Platz, M. (2017). Digital platforms for research collaboration: Using design science in developing a South African open knowledge repository. *Information Technology for Development*, *23*(3), 463–485. https://doi.org/10.1080/02681102.2017.1328654

Waldfogel, J. (2017). How Digitization Has Created a Golden Age of Music, Movies, Books, and Television. *Journal of Economic Perspectives*, *31*(3), 195–214. https://doi.org/10.1257/jep.31.3.195

Watzinger, M., Fackler, T. A., Nagler, M., & Schnitzer, M. (2020). How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree. *American Economic Journal: Economic Policy*, *12*(4), 328–359. https://doi.org/10.1257/pol.20190086

Whetsell, T. A., Dimand, A. M., Jonkers, K., Baas, J., & Wagner, C. S. (2021). Democracy, Complexity, and Science: Exploring Structural Sources of National Scientific Performance. *Science and Public Policy*, *48*(5), 697–711. https://doi.org/10.1093/scipol/scab036

Winkler, A. E., Levin, S. G., & Stephan, P. E. (2010). The diffusion of IT in higher education: Publishing productivity of academic life scientists. *Economics of Innovation and New Technology*, *19*(5), 481–503. https://doi.org/10.1080/10438590903434844

Wittel, A. (2013). Counter-commodification: The economy of contribution in the digital commons. *Culture and Organization*, *19*(4), 314–331. https://doi.org/10.1080/14759551.2013.827422

World Bank. (2018). *Information and Communication for Development 2018: Data-Driven Development*. The World Bank Group. http://documents.worldbank.org/curated/en/987471542742554246

Xie, X., Gao, Y., Zang, Z., & Meng, X. (2020). Collaborative ties and ambidextrous innovation: Insights from internal and external knowledge acquisition. *Industry and Innovation*, *27*(3), 285–310. https://doi.org/10.1080/13662716.2019.1633909

Yu, Y., Ibarra, J. E., Kumar, K., & Chergarova, V. (2021). Coevolution of cyberinfrastructure development and scientific progress. *Technovation*, *100*, 102180. https://doi.org/10.1016/j.technovation.2020.102180

Zellner, A. (1962). An efficient method of estimating seemingly unrelated regressions and tests for aggregation Bias. *Journal of the American Statistical Association*, *57*(298). https://doi.org/10.1080/01621459.1962.10480664

Zhang, G., Wang, Y., Xie, W., Du, H., Jiang, C., & Wang, X. (2021). The open access usage advantage: A temporal and spatial analysis. *Scientometrics*, *126*(7), 6187–6199. https://doi.org/10.1007/s11192-020-03836-4

Zhang, G., Wang, L., Shang, F., & Wang, X. (2025). What are the digital skills sought by scientific employers in potential candidates? *Journal of Higher Education Policy and Management*, *47*(1), 20–37. https://doi.org/10.1080/1360080X.2024.2374392

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